RealQM on Conduction

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Abstract

This article computes the activation energy for charge transport in a solid from Coulomb interaction and kinetic energy alone, with nothing fitted. No such computation appears to have been carried out before: the equation that determines the quantity from those two terms is the $3N$-dimensional Schrödinger equation, which cannot be solved, and every route that can be followed introduces something further — a chosen functional, a relaxation time, a transfer integral. The framework used is RealQM, a new *ab initio* alternative to textbook quantum mechanics. RealQM represents an N-electron system in terms of N real-valued wave functions $\psi_i(x)$, with x a coordinate in three-dimensional Euclidean space, whose squares ψ_i^2 are unit charge densities with non-overlapping supports meeting at free interfaces. The physics of the system is described through its energy functional, the sum of electron Coulomb potential energy without self-repulsion and kinetic energy: no other force enters it, and nothing in it is fitted. The ground state is the configuration minimising that functional over the supports as well as over the wave functions, so the interfaces are unknowns of the problem rather than inputs to it, and the cost is polynomial of small power in N.

Computationally the difficulty is one of scale. Atomic energies are tens of electron-volts, semiconductor activation is 50–500 meV per carrier and metallic conduction sets in near 20 meV, while the total energy of a chain long enough to pose the question is thousands of electron-volts. The barrier sought is therefore some 10^{-5} of the number a calculation returns. It is not reached by computing that number to five figures, but by differencing two configurations of the same system — the same electrons, the same kernels, the same grid, differing only in where a few interior domain walls sit — so that every large contribution to the error is common to both and cancels.

Three arrangements of charge are computed, and the barrier to advancing the electron pattern by one site differs by more than an order of magnitude between them. These are not one-dimensional models. The wave functions are functions on ordinary three-dimensional space, the domains are regions of it, and the calculations run on grids of some two to three million points; what is arranged along a line is the nuclei. A **chain of hydrogen atoms**, the electron on its own kernel, gives ≈580 meV per electron and a polarisability $\alpha = 7.4N$, extensive in the number of electrons. A **cable of Li ions**, the carriers standing off closed cores, gives ≈230 meV. The same cable with the carriers stiffened — the coefficient of the kinetic energy doubled — gives ≈31 meV, and with the sign inverted: the preferred registry moves from the carriers sitting on their ion sites to sitting between them.

The barrier is the activation energy in a hopping conductivity $\sigma = (ne^2a^2\nu/kT)\, e^{-E_a/kT}$. Closing the prefactor with an attempt frequency $\nu = (1/2\pi)\sqrt{K/M}$ built from the curvature of the same landscape and a nuclear mass gives ν of 92 and 22 meV — lattice frequencies, not asked for — and room-temperature conductivities of 1×10^{-4} and 19 $\mathrm{S\,m^{-1}}$, comparable to intrinsic silicon and to intrinsic germanium. For the stiffened cable the expression does not apply, a barrier at $1.2\,kT$ being no barrier to speak of. What the framework supplies is the exponent; the nuclear mass and the carrier density are part of specifying the system and of reporting it,

not parameters fitted to anything: nothing in the model is adjusted to reproduce a measured number. The barriers are upper bounds, since charge moves by kinks rather than rigid slides. A resistive conductivity is outside altogether, needing a relaxation time from electron–phonon scattering.

1 The basic question posed to RealQM

This article computes the activation energy for charge transport in a solid from Coulomb interaction and kinetic energy alone, with nothing fitted — a computation that does not appear to have been carried out before, for reasons §2 sets out. The framework is RealQM, a new *ab initio* alternative to textbook quantum mechanics.

RealQM represents an N-electron system in terms of N real-valued wave functions $\psi_i(x)$, with x a coordinate in three-dimensional Euclidean space, whose squares ψ_i^2 are unit charge densities with non-overlapping supports Ω_i meeting at free interfaces. The physics of the system is described through its energy functional, the sum of electron Coulomb potential energy without self-repulsion and kinetic energy, with no other force entering and nothing fitted. Explicitly,

$$E[\{\psi_i\},\{\Omega_i\}] = \sum_i \int_{\Omega_i} \Big(\tfrac{1}{2}|\nabla\psi_i|^2 - \sum_s \frac{Z_s}{|x-x_s|}\psi_i^2\Big)dx + \tfrac{1}{2}\sum_{i\neq j}\iint \frac{\psi_i^2(x)\psi_j^2(y)}{|x-y|}\,dx\,dy, \tag{1}$$

where i and j index the electrons, s indexes the nuclei, Z_s and x_s are the charge and position of nucleus s, and y is a second space coordinate; the supports are disjoint and $\int_{\Omega_i}\psi_i^2 = 1$. Atomic units are used throughout: lengths in Bohr radii $a_0 = 0.529$ Å, energies in hartree, 1 Ha = 27.211 eV.

The ground state of the system is the configuration minimising this energy, the minimisation running over the supports as well as over the wave functions, so that the interfaces are unknowns of the problem and not inputs to it; excited states are the other stationary points of the same energy. The cost of minimising (1) is polynomial of small power in N.

Every previous test of RealQM — ionisation energies across the periodic table [21], molecular binding, hydrogen bonds and solvation [18, 20], nuclear binding by dual Coulomb confinement [22] — has been on *bound* matter, where an attractor is always present and the charge is where the attractor is. Conduction is the first question that asks about charge which is not where an attractor holds it, and the framework has no separate machinery for it: the same energy, the same domains, in a different arrangement of the same two kinds of charge.

Three-dimensional throughout. Both arrangements below place their nuclei on a line, and it is worth saying at once that nothing else about them is one-dimensional. Each ψ_i is a function on $\mathbb{R}^3$, each support Ω_i is a region of it with a two-dimensional free surface, and the domains tile a three-dimensional box — 138^3 and 148^3 points for the two systems here. A linear chain of nuclei is the smallest arrangement in which a registry between a carrier lattice and an ion lattice can be posed at all, which is why it is used; it is not a reduction of the electronic problem, and there is no step anywhere in this article at which a dimension is integrated out.

Two arrangements, both ordinary matter. The first is a **chain of atoms**: one kernel per site with its valence electron occupying the same region, which is what a hydrogen chain is and what an alkali is to a first approximation. The second is a **cable of ions surrounded by electrons**: kernels screened by their own core electrons, with the valence charge standing off outside them, which is how a wire is built. The question is what each gives.

2 What textbook quantum mechanics offers

Recall that textbook quantum mechanics is based on Schrödinger's equation in $3N$ dimensions for a single antisymmetric wave function $\Psi(x_1, \ldots, x_N)$, with the exclusion principle carried by the antisymmetry and with a Hamiltonian built, like (1), from Coulomb interaction and kinetic energy alone. But the cost of solving it is exponential in N, and the models actually used — Hartree, Kohn–Sham and their descendants — always include drastic reductions of the dimensionality. There is thus no textbook treatment of conduction and insulation based on Schrödinger's equation. There is a stack of models, each carrying quantities supplied from measurement.

Metal is separated from insulator by a gap at the Fermi level [1], and in practice the gap comes from a chosen exchange–correlation functional [2]; the Kohn–Sham gap is not the excitation gap even for the exact functional [3], and the discrepancy is repaired by hybrid mixing or a GW self-energy [4]. Transport is then a second model on top of the first: a Boltzmann equation closed by a relaxation time, with an effective mass and a deformation potential for the semiconductor [5]; a transfer integral and a reorganisation energy for a hopping conductor [6, 7]; a localisation length and a density of states at the Fermi level for variable-range hopping in a disordered one [8]; a Hubbard U where the band picture fails outright and the material insulates by correlation instead [9, 10]. The recent direction of travel is worth noting, because it runs the other way: learned exchange–correlation functionals [11] replace a handful of constructed parameters with a neural network trained on reference data, and interatomic potentials fitted to density-functional output [12] dispense with the functional altogether in favour of a model fitted end to end. These are effective, and the neural wavefunction methods [13] are a genuine advance on the variational problem — but they solve the $3N$-dimensional problem with a better ansatz, and the density-functional line answers the cost of that problem by adding fitted content rather than removing it.

The point of the contrast is not accuracy — these are accurate, and are what one should use to predict a number for a given material. It is that they are several models with parameters, addressing the regimes separately, whereas what follows is one energy with none, in which they differ by how the charge is arranged.

How the comparison should be read. Two misreadings are worth heading off, because they pull in opposite directions. The first is to score RealQM against the accuracy of methods that carry fitted quantities. That is the wrong axis: the claim here concerns what the model contains, not whether it beats a percentage, and a framework with no adjustable content should not be judged by one that has some.

The second is to require agreement with a near-exact solution of Schrödinger's equation — configuration interaction, coupled cluster, quantum Monte Carlo — for the same system. RealQM is not an approximation to that equation. It is a different formulation, and it is not empirically equivalent to it: the two make different predictions, and where they differ the question is which the world agrees with, not which reproduces the other. Disagreement with a benchmark solution of the $3N$-dimensional problem is therefore not a defect in a formulation that does not claim to solve it.

What remains, and it is the only thing that adjudicates, is measurement. RealQM's record there is on bound matter: the binding of H_2 comes out 4.20 eV against 4.75 measured, and ionisation energies across the periodic table are reproduced without fitting [21, 18].

For transport there is as yet nothing, and it is worth being exact about why. The systems computed below are model systems: a chain of eight hydrogen atoms and a cable of sixteen lithium ions, chosen because they are the smallest arrangements in which the question can be posed at all. Neither has an experimental counterpart — nobody has measured the conduction barrier of a linear H_8 chain — so the absence of a comparison is a consequence of what was computed, not a shortcoming of the framework that computed it. Model systems without counterparts are ordinary; the Ising model and jellium are studied for what they isolate, not for what they reproduce.

The consequence is to be accepted rather than dressed. Conductivities of real materials appear below only to set a scale, and no agreement with them is claimed or implied. What would supply a comparison is the activation energy, which is exactly what an Arrhenius plot of a real activated conductor measures, computed for a system that has one. That is a matter of choosing a larger and more realistic cell, which the cost of the method permits; it is not a matter of the method improving.

Equation (1) has no $3N$-dimensional object in it and no antisymmetry: the role played by the exclusion principle is played instead by the requirement that the supports do not overlap.

3 What RealQM computes

A domain carries one unit of charge by constraint, so no charge passes between sites. What can move is the *partition*: the boundaries are redrawn, and the pattern of domains advances through the lattice. When the pattern has advanced by one full spacing, every electron has taken over its neighbour's kernel and the configuration is restored — identical, since the electrons are identical and the energy depends on the partition and not on labels. Charge has been transported and nothing material has traversed the sample, which is the relation a dislocation bears to slip.

The quantity that decides whether this happens is the energy along that path. Write $E(d)$ for the energy with the partition displaced by a fraction d of a spacing. It is periodic, $E(d+1) = E(d)$, and its amplitude

$$V_0 = \max_d E(d) - \min_d E(d) \tag{2}$$

is the corrugation: the barrier the partition must cross to advance one site. A field tilts this curve by $-NeFd$, so the pattern runs when the tilt exceeds the steepest slope of $E(d)$, and otherwise settles at a displacement and stops — bounded polarisation, no current.

Ends, and how to remove them. A finite chain has ends, and they dominate: terminal domains bind an electron-volt harder than interior ones, so a corrugation read from total energies is reading the ends. Displacing only the middle W carriers repairs this. The energy difference is then W times the bulk cost plus two junction costs, so differencing two window sizes cancels the junctions exactly, and no terminal domain moves in either configuration:

$$\Delta E(W) = W\,\varepsilon_{\mathrm{bulk}} + 2\varepsilon_{\mathrm{junc}}, \qquad \varepsilon_{\mathrm{bulk}} = \frac{\Delta E(W_2) - \Delta E(W_1)}{W_2 - W_1}. \tag{3}$$

The decomposition asserts linearity in W, which a third window tests rather than fits.

Why tens of meV are available out of a hundred hartree. The two are not the same kind of quantity, and it is worth separating them before the objection is answered. The hartrees are the total energy of the whole system — forty-eight electrons on sixteen sites — whereas the barrier is quoted *per carrier*. What is actually differenced is $\Delta E(W_2) - \Delta E(W_1)$, of order 3×10^{-2} Ha, and dividing by $W_2 - W_1$ turns it into the per-carrier figure. So the quantity to be resolved is about 2×10^{-4} of the total rather than 10^{-6}.

That is still four significant figures down, and the natural objection is that no calculation resolves it. The objection would be right if the total energy had to be accurate. It does not. What is required is that the *difference* be accurate, and the two configurations entering it are the same system: the same domains, the same electrons, the same kernels, the same total charge, on the same grid, differing only in where a few interior walls sit. Every large contribution to the discretisation error — the nuclear regions, the cores, the box — is identical in both and subtracts out. This is the ordinary reason binding energies are quoted in meV while the total energies they come from carry errors a thousand times larger.

That the cancellation actually occurs is not assumed but checked, in four independent ways. The solver is deterministic: repeated runs of one configuration agree to the last reported digit, so there is no stochastic floor beneath the quantity being computed. Translation by one full period is an exact relabelling and must return the same energy, which it does to nine digits — two separately relaxed states agreeing to that precision. The junction cancellation in (3) is exact algebra rather than an approximation, so the instrument introduces no error of its own. And the linearity the decomposition asserts is tested against a third window, which it could fail and does not. What the last of these does not do is prove the result correct; symmetry controls verify self-consistency, and a systematic error respecting the lattice symmetry would pass all of them. They establish that the arithmetic is sound, which is a smaller claim than the result being right, and a necessary one.

4 A chain of atoms

Eight hydrogen atoms in a row at spacing $a = 4.5\,a_0$: kernel charge +1, one electron per site, $r_c = 0$. There is no core, so there is no radius to impose and nothing excluded by a rule — each electron is kept off its neighbours by non-overlap alone. This is the simplest arrangement in which the carrier occupies its own kernel's region, and it is a real system rather than a model atom.

The barrier. Sliding the whole pattern by half a spacing costs

$$\varepsilon \approx 580 \text{ meV per electron.}$$

The control on this is exact: on a wrapped chain a displacement of one full spacing is a relabelling and must return the same energy, and two independently relaxed runs give $E(d{=}0) = E(d{=}1) = -3.624632386$ Ha to all nine digits. The result also proves insensitive to the one arbitrary length in it — repeating with the nuclear cutoff at $0.8\,a_0$ instead of $0.6\,a_0$ moves the total energy by 6.3 eV and the barrier by 0.9 meV, which is the error cancellation of §3 shown rather than argued.

The polarisability is extensive. With the partition frozen and a uniform field applied, the induced dipole gives α, and repeating at three chain lengths gives its scaling:

$$\alpha = 7.4\,N\,a_0^3, \qquad \text{flat to 2.5\% across } N = 4, 6, 8.$$

An isolated hydrogen atom has $\alpha = 4.5\,a_0^3$, so the per-atom value is a modest chain enhancement of it. The scaling is what matters. A response strictly proportional to N is the signature of charge that cannot leave its site: each domain carries its unit by constraint, so the only responses are the density deforming within a fixed domain and the boundary displacing, both per-atom constants. Charge free to redistribute across the sample would give a response growing faster than N, and does not.

5 A cable of ions

The chain puts the carrier where the attractor is. Conducting matter does not: core electrons occupy the region around each nucleus and the conduction charge stands outside them. Representing that needs two populations, and it is the one arrangement giving a carrier a standoff without violating the tiling, because the inner region is owned rather than left empty.

Sixteen sites at $a = 4.5\,a_0$, kernels of charge $+3$, a closed core of two electrons at each, one valence electron per site outside them: neutral, and $r_c = 0$ — the core is represented by its electrons rather than by a pseudopotential standing in for them, so the carrier is kept out of the core region by non-overlap and not by an imposed condition. The core pair is a single homogenised domain of charge two rather than two unit domains meeting at a plane — a closed shell is spherically symmetric, and split halves relax unequally, which the eigenvalue control catches. Its self-repulsion is reduced to match: for a homogenised domain of charge q the fraction $1/q$ of its own potential is removed, leaving $(q-1)/q$ of the naive self-energy, exactly the $\binom{q}{2}$ pairs the q electrons form. At $q = 2$ that is one pair; at $q = 1$ the self-potential goes entirely, recovering the usual absence of self-interaction. The carriers are $q = 1$ throughout.

The barrier, evaluated by (3) with windows of $W = 4$ and $W = 8$ carriers, is

$$\varepsilon_{\text{bulk}} \approx 230 \text{ meV per carrier},$$

a factor of 2.5 below the chain. The interaction is the same and the energy functional is the same; what has changed is where the carrier sits relative to its kernel.

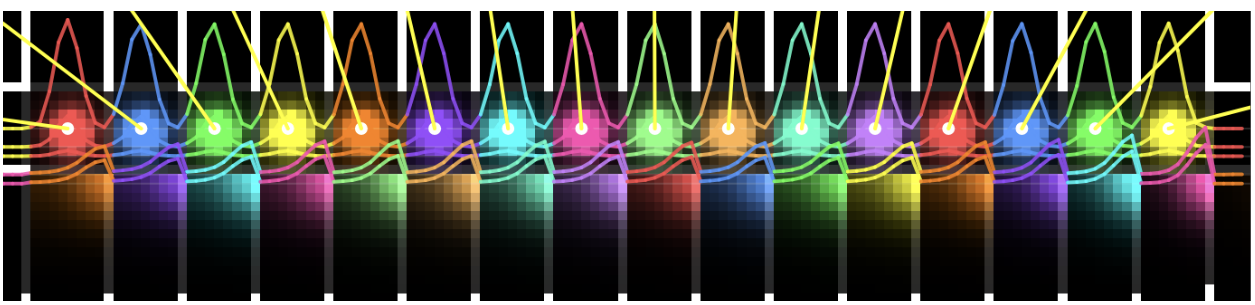

Figure 1: The cable, relaxed. Sixteen sites along the axis, each carrying a kernel of charge $+3$, a core domain of two electrons confined within $c_r = 2.4\,a_0$, and one valence electron outside it. Colour distinguishes domains: the bright lobes on the axis are the cores, the fainter blocks beneath them the valence charge standing off. White verticals mark the free boundaries between neighbouring carriers, and the coloured curves are density profiles along a line through the axis; the yellow segments are net forces on the kernels. Every boundary shown is an unknown of the minimisation, not an imposed partition. Displacing the middle M of these carriers by half a spacing, at fixed cores and kernels, is the quantity computed in §3.

The standoff is imposed, and narrow. The radius c_r separating cores from carriers is prescribed, not found. It is a constraint on the minimisation and an expensive one: at fixed particle content the energy falls steeply as c_r is reduced, by of order ten electron-volts per electron over the range examined, so the configuration is held open against a strong preference to collapse. Released — the same system with that interface free — the carrier shell falls onto the axis, and every starting standoff returns the same energy. Nor can the barrier be traced along c_r as a free coordinate, at least downward: at this mesh a smaller radius squeezes the core between the nuclear cutoff and its own boundary, leaving it in a nearly flat well that does not relax, so the atom-like end of the range is not reachable here. The cable is therefore a stated arrangement whose conduction is computed, not a configuration RealQM produces of itself, and the dependence of the barrier on standoff — which is the mechanism proposed above — is not demonstrated here.

6 What sets the pinning

The barrier is not set by the strength of the ion lattice but by whether the carriers bunch into its wells. A carrier distribution that resisted being modulated would ride over the lattice; one that does not resist locks to it.

RealQM's free interface resists very little, and this is internal to the construction. A flat density is admissible on a free boundary at no cost — a constant ψ_i satisfies the normalisation, has $\nabla\psi_i = 0$, and minimises the electrostatic energy against a neutralising background — so redrawing an interface between domains of equal density costs nothing. The partition therefore offers no resistance to uniform compression, and the only penalty on a *modulated* density is the gradient term in (1), carrying the coefficient $\frac{1}{2}$.

That coefficient is a standardisation, not a constant of the theory. It is set by calibrating the framework against atomic spectra, hydrogen first among them — on a bound electron in a single potential well. An extended arrangement is a different object, as an effective mass in a solid differs from the free-electron mass without anyone regarding it as tampering, and the coefficient governing how a carrier lattice resists modulation by an ion lattice has no obligation to equal the one an atom's spectrum sets.

So the natural thing to ask is how much the answer turns on it, and that is all the following is: a sensitivity, not a calibration. The coefficient is doubled — a round factor, chosen for being round — and the corrugation is computed again. No value was sought that would reproduce a conductivity, a barrier, or any other quantity taken from experiment, and none of the numbers below was known in advance. Doubled, with the ions, the standoff, the geometry and the mesh unchanged:

$$\varepsilon_{\text{bulk}} \approx -31 \text{ meV per carrier},$$

and the sign is the result. A negative value does not mean the barrier has gone: on a wrapped chain $d = 1$ *is* $d = 0$, so a preference for the displaced configuration means the minimum has moved to $d = \frac{1}{2}$ and the maximum now sits at $d = 0$. The barrier between successive minima is $E(0) - E(\frac{1}{2}) \approx 31$ meV per carrier, a fall by a factor of 7.3 from the unstiffened cable, and the preferred registry has inverted — from carriers sitting on their ion sites to carriers sitting between them.

That is what a stiff carrier lattice should do. A lattice that resists being modulated resists being pulled into the ion wells, and past some stiffness it prefers the configuration that keeps its own spacing uniform. The charge is not thereby free: 31 meV is $1.2\,kT$, small but not absent, and it still holds the carriers — in a different place. The response rests on two values of the coefficient rather than a curve, and on two endpoints $d = 0$ and $d = \frac{1}{2}$, so that the extrema lie exactly there is an assumption the sign inversion makes worth testing with an intermediate point.

arrangement	barrier per carrier	preferred registry
chain of H atoms, carrier on its kernel	580 meV	carrier on its site
cable of Li ions, carrier standing off	230 meV	carrier on its site
the same cable, carriers stiffened	31 meV	carrier between sites

7 What this gives, and what it does not

In the activated regime the corrugation is the activation energy, and the conductivity of a hopping conductor is

$$\sigma = \frac{ne^2a^2\nu}{kT}\, e^{-E_a/kT}, \qquad \nu = \frac{1}{2\pi}\sqrt{\frac{K}{M}}, \tag{4}$$

with σ the conductivity, n the carrier density, e the elementary charge, a the hop distance, E_a the activation energy, k Boltzmann's constant, T the temperature, ν the attempt frequency, K the curvature of the well the carrier sits in and M the mass of whatever oscillates in it.

Activated rate theory. The picture behind (4) is Arrhenius's, made quantitative [14]. A system sits in a potential well separated from the next by a barrier E_a. It does not cross steadily. It oscillates in the well at the small-oscillation frequency set by the curvature of the well and the mass in it, and on each oscillation there is some chance that thermal fluctuation has put enough energy into the right coordinate for it to get over. The rate is therefore a product of how often it tries and how often a try succeeds: ν counts the attempts, and $e^{-E_a/kT}$ is the fraction of the time the Boltzmann distribution holds at least E_a there.

The theory carries two conditions, and one of them will matter below. The barrier must be large against kT: if it is not, crossings are not rare, the system spends a substantial part of its time above the top, and the rate is set by how fast it moves rather than by how seldom it is lucky — so the factorisation into attempts and successes stops describing anything. And the system must lose the memory of one attempt before making the next, which in a solid it does by exchanging energy with the lattice.

What oscillates here is the collective coordinate u: the displacement of the carrier pattern along the chain, the same variable $E(u)$ is evaluated against in §3. The attempt is therefore the charge pattern rocking back and forth in one corrugation well against the ion lattice, and since it is the sites that move, that rocking is a lattice vibration. Both V_0 and M are then taken per carrier, which is consistent: displacing m carriers rigidly costs m times the barrier and moves m times the mass, and ν is unchanged.

RealQM supplies E_a, which decides the temperature dependence and the ratio between arrangements. The prefactor needs K, M and n, and it is worth being exact about where each comes from.

K is the curvature of the landscape whose amplitude was computed. Taking the periodic curve to be $E(u) = \frac{1}{2}V_0\big[1 - \cos(2\pi u/a)\big]$ gives $K = 2\pi^2 V_0/a^2$, which for $V_0 = 230$ meV at $a = 4.5\,a_0$ is 0.80 eV Å^{-2}. This is the one step that assumes a shape rather than measuring one: the windowed slide was run at the two endpoints $d = 0$ and $d = \frac{1}{2}$, which fix the amplitude and not the curvature. A scan at small d would replace the assumption, and until it is run the sinusoid stands in for it.

M is part of the specification of the system rather than a parameter of the theory. Two masses must be kept apart here. The *electron* mass is genuinely absent from (1): the coefficient of $|\nabla\psi_i|^2$ is a geometric scale, not $\hbar^2/2m$, which is why §6 can vary it and why E_a comes out without any mass being supplied. The *nuclear* mass is a different kind of thing — a property of the nuclei one has chosen to put in the box. To say the material is lithium is already to say $Z = 3$ and $M = 7\,u$, exactly as fixing the nuclear positions says what the lattice is. It is needed only when the question turns from what the energy is to how fast something oscillates in it. What moves here is the site, since the charge does not pass between sites but the pattern of domains advances through the lattice, so M is a nuclear mass and ν comes out a lattice frequency.

$n = 1/a^3$ is a convention for reporting rather than a quantity of either kind. It takes the cable's spacing to hold in three dimensions, one carrier per site. The cable is a single line of sites, so this is a stated convention for converting a per-carrier barrier into a bulk conductivity, not something the calculation contains.

With the nuclear mass of the moving species — $1\,u$ for the hydrogen chain, $7\,u$ for the Li cable — and $T = 300$ K:

	E_a	ν (Hz)	σ (S/m)	for scale
chain of H atoms	580 meV	2.2×10^{13}	1.0×10^{-4}	intrinsic Si, 4×10^{-4}
cable of Li ions	230 meV	5.3×10^{12}	19	intrinsic Ge, 2
cable, stiffened	31 meV	1.9×10^{12}	(1.5×10^{4})	— see below

The attempt frequencies are 10^{12}–10^{13} Hz, which is where lattice vibrations live: 22 meV for the cable and 92 meV for the hydrogen chain, the latter high because a proton is light. Nothing was arranged to make that so, and it is the check that the mass entering ν is the right kind of mass. The first two rows bracket the conductivity of ordinary semiconducting material, three orders of magnitude apart on a barrier ratio of 2.5.

The third is bracketed because it violates the first condition above. At $E_a = 31$ meV against $kT = 25.9$ meV the barrier is $1.2\,kT$ and $e^{-E_a/kT} = 0.30$: the carriers are over the top nearly a third of the time, so escape is not rare and ν is not counting attempts. The number is what (4) returns when applied outside its own regime, and should be read as a marker that the barrier has become small rather than as a conductivity. What rate law governs that state, and what the state is, this calculation does not determine.

These are order-of-magnitude statements and cannot be more. The ± 30 meV on E_a is already a factor 3.2 either way in σ at room temperature, the sinusoid is an assumption, and M and n are inputs. What the framework delivers is the exponent; that the resulting numbers land in the three named regimes, in the right order, with nothing fitted, is the result.

The kink, and why the table is a lower bound. Every barrier above is a rigid slide: all carriers displaced together, so the cost is N times a per-carrier barrier and the whole sample must climb at once. Real conductors do not do this. The cheap path is a kink — one carrier advancing while the rest hold still, the domains ahead of it compressed and those behind expanded — a localised defect whose energy does not grow with the length of the chain and which, once formed, moves by shifting its centre through the lattice. This is the relation a dislocation bears to sliding a whole crystal plane, and it is the same object a charge-density wave transports charge with [17].

The carriers-on-a-substrate system of §5 is a Frenkel–Kontorova chain [15]: a harmonic coupling κ between neighbouring carriers on the periodic potential of curvature K computed here. Whether such a chain pins at all is the Aubry transition [16], controlled by the same ratio of the two constants. Two consequences follow, and both bear on the numbers above. First, the barrier changes character. A kink of width $w = \sqrt{\kappa/K}$ sites crosses a Peierls–Nabarro barrier E_{PN} which falls steeply — exponentially, in the continuum limit — as w grows, so a wide kink slides almost freely over a substrate that pins a rigid chain hard. The conductivity is then governed by $E_f + E_{\mathrm{PN}}$, the cost of creating a kink plus the cost of moving it, in place of E_a; since $E_{\mathrm{PN}} \ll V_0$ the exponent falls and every σ in the table is an underestimate, by a factor this article does not determine.

Second, and to the question of a clock: the kink is better supplied with one than the rigid slide is. The same two constants give the carrier lattice a sound speed $c_0 = a\sqrt{\kappa/M}$, which is the kink's limiting velocity, and a kink mass set by κ, K and M — so the rate of kink transport is fixed by κ, K, E_f, E_{PN} and M, of which only M comes from outside. The width w in particular is a pure ratio of two curvatures and therefore wholly a RealQM quantity. Nothing new in kind is needed; what is needed is κ, which is the curvature of the energy against a *non-uniform* displacement of the carriers, and E_{PN}, which must be inferred from κ and K rather than read off a slide. Neither is computed here.

What doping would be. This is where the identification with the activated conductors earns or loses its keep, because those materials are doped, and in this framework doping cannot mean what it usually means. There is no band to add a carrier to and no gap to promote one across. What can be added is a *domain* — and since the domains tile, an added domain takes its space from its neighbours and the carrier lattice is locally compressed, while a removed one leaves the neighbours to expand and it is locally stretched. Those are the kink and the antikink. So n- and p-type would be compression and rarefaction of the carrier lattice, dopant concentration would set the density of kinks, and conduction would proceed by moving the very defects that doping creates.

Two consequences are worth stating because they are checkable in principle. The activation energy of doped material would be E_{PN} rather than the rigid-slide E_a, and $E_{\mathrm{PN}} \ll V_0$ — which is the qualitative fact that doped silicon activates at tens of meV where the gap is over an electron-volt, arrived at here without a gap existing. And the asymmetry between n- and p-type would be structural rather than fitted, since a kink and an antikink cross different Peierls–Nabarro barriers on a substrate that is not symmetric about its minimum.

None of this is computed. It requires E_f and E_{PN}, which as above are not computed here, and until they are it is an entailment of the construction rather than a result of it.

What is outside is not a regime but a kind of quantity. A resistive conductivity $\sigma = ne^2\tau/m$ needs a momentum relaxation time from electron–phonon scattering, and that needs a current as an object, electron dynamics, and irreversibility; a minimisation over real densities has none of the three, at any barrier height. And band conduction specifically is absent in kind rather than by degree: non-overlap admits no extended states, so there is nothing for a band to be made of, and no choice of stiffness produces one. Neither of these tells us what the unpinned state of §5 is. They say only that if it conducts, it does not do so by filling bands, and that this framework will not supply the rate at which it does.

8 Further studies

The nuclei reported here lie on a line. The calculation does not: it is three-dimensional throughout, and what is restricted is the lattice, not the electronic problem. A line is the smallest arrangement in which the question can be posed, and it is why $n = 1/a^3$ had to be a convention rather than a count. The obvious next step is real configurations in three dimensions: the tetrahedral lattice of silicon and germanium, the close-packed lattices of the alkali metals, the layered structures of the oxides. Nothing in (1) is one-dimensional, and the angular splitting of a kernel's valence sector already reaches the bonding topologies these lattices are built from. A three-dimensional cell would make the carrier density a property of the structure rather than a stated convention, and would let a barrier be compared with the activation energy of a named material instead of with a range.

The scale needed for this is not out of reach, and that is the practical consequence of a cost polynomial of small power in N. Solids have already been computed in this framework for structural and thermal questions — ionic crystals under melting and under shear, molecular solids including ice, solid nitrogen and carbon dioxide, and a lithium metal lattice of a hundred atoms restoring itself from a perturbation under dynamics [19]. What has not been attempted in any of them is transport. It is also what several of the questions below require rather than merely benefit from. A dopant at a realistic concentration is one atom in 10^3 or fewer, so doping cannot be represented on sixteen sites at all — the account of n- and p-type in §7 needs a cell large enough to hold one dopant and the lattice around it. The same is true of a kink, whose width is set by the ratio of two curvatures and which must fit inside the cell with bulk on either side of it, and of grain boundaries, interfaces and vacancies, which are where the transport of real material is usually decided.

The one quantity still owed can be discharged two ways. A scan of $E(d)$ at small d gives the curvature directly and retires the assumed sinusoid. Better, moving the nuclei under their classical equations of motion — which the framework does, and which is the one genuine clock in it, the relaxation iteration being imaginary time — gives the attempt frequency as a period read off a trajectory, making both the curvature and the assumed shape unnecessary.

The kink is the larger prize. Every barrier here is a rigid slide, so every barrier is an upper bound and every conductivity a lower bound by a factor this article does not determine. Measuring the formation energy E_f and the Peierls–Nabarro barrier E_{PN} would replace bounds with values, and would at the same time make the account of doping in §7 testable: n- and p-type as compression and rarefaction of the carrier lattice, dopant concentration setting the kink density, and an n/p asymmetry that is structural rather than fitted.

One defect in the discretisation blocks more than any single calculation. Positions live on the grid — a domain wall sits in one cell or the next, and nuclear coordinates are truncated to whole cells when they are handed to the solver — so a displacement smaller than a cell does not exist. At the mesh used here one cell is an eighth of a spacing. Six quantities are held up by this one limitation, and they have the appearance of separate difficulties only because they were met separately. The polarisability must be computed with the partition frozen, since a released boundary answers a sub-threshold field by moving zero cells or one. A boundary driven at fixed force either sticks or chatters between adjacent cells, so a depinning threshold cannot be read. The Peierls–Nabarro barrier needs a kink profile finer than a cell. And κ, the second Frenkel–Kontorova constant, requires a non-uniform displacement small enough to be harmonic — smaller than the grid can express, so amplitudes below a cell return the undisplaced chain exactly.

The same limitation reaches the nuclei, and two further things follow. The attempt frequency was to be obtained by releasing the kernels and reading a period off a trajectory, which would retire the assumed curvature; but a thermal oscillation at room temperature has amplitude around $0.3\,a_0$, under a cell at this mesh, so the nuclei would not move. And a conjugated chain such as polyacetylene distorts by alternating its bonds — the Peierls instability that makes it a semiconductor and gives its solitons something to be a defect in — by about $0.03\,a_0$ per carbon, a sixth of a cell. Neither is a matter of longer runs or a larger machine. Carrying positions as continuous coordinates rather than cell indices, and the boundary as a volume fraction rather than by whole-cell ownership, would release all six.

That constant is worth naming as an aim in its own right. The chain of carriers on a periodic substrate is a Frenkel–Kontorova system [15], the standard phenomenological account of commensurate lattices, depinning and kink transport, and its two constants are the substrate amplitude and the coupling. RealQM computes the first: V_0 is the corrugation computed here, which in ordinary use is supplied from data. With the second, the same model yields the kink width, the Peierls–Nabarro barrier and the Aubry pinning criterion [16] with nothing further put in. The transport would then be done by a model whose credentials are not in question, from constants this framework computed rather than fitted — which is a sharper claim than a conductivity resting on an assumed curvature. It is worth noting which standard models are *not* available in the same way: a Hubbard or tight-binding description is degenerate here, since non-overlap makes the transfer integral vanish identically, and Frenkel–Kontorova fits because its degrees of freedom are the ones RealQM has.

Two questions are open in a way that could change the account rather than sharpen it. Whether RealQM has a self-consistent standoff — a cable the functional chooses rather than one held open against a strong preference to collapse — is the first, and §5 records that released, it does not. The second is whether the arrangement in which the barrier falls to thermal energy conducts in any useful sense, or is merely a lattice in a different registry; the polarisability scaling is the test, since charge fixed per site gives $\alpha \propto N$ whatever the barrier.

Finally, nothing here has been compared with a measured conductivity. The framework's record is on bound matter, and transport is a different regime. The sharpest available test is not a value but a slope: the activation energy fixes the temperature dependence of σ, and an Arrhenius plot of a real activated conductor measures the same quantity directly.

9 Conclusion

For a chain of hydrogen atoms, the electron occupying its own kernel's region, RealQM gives a barrier of 580 meV per electron for the electron pattern to advance one site, and a polarisability $\alpha = 7.4N$, extensive in the number of electrons.

For a cable of Li ions with the carriers standing off closed cores it gives 230 meV, a factor of 2.5 less. The two cannot be built from the same ion, and the reason is instructive: a carrier can stand off a core only if there is a core, and since the domains tile, one domain per kernel must contain its kernel — so a standoff requires at least two electrons per site where the chain has one. The difference in ion is entailed by the difference in arrangement rather than chosen alongside it. What is the same is the physics: (1) throughout, with nothing fitted in either.

Stiffening the carriers of that cable gives 31 meV, and inverts the preferred registry: the carriers come to prefer sitting between the ion sites rather than on them. So the barrier is set less by the ion lattice than by how strongly the carrier distribution resists being modulated by it, which is a property of the interface condition rather than of the interaction.

Taking each barrier as an activation energy and closing the prefactor with an attempt frequency built from the same landscape and a nuclear mass puts the first two arrangements at 10^{-4} and $10^{1}\ \mathrm{S\,m^{-1}}$ — intrinsic silicon and intrinsic germanium — with the attempt frequencies coming out at lattice values unasked. Nothing in this is fitted. What the calculation takes beyond (1) is a specification of the system — which nuclei, at what spacing, under what geometric constraint — together with a convention for reporting a bulk density; and one quantity, the curvature, which is a property of the computed landscape that has been assumed rather than computed, since only the endpoints of $E(d)$ were run. The conductivities are lower bounds, since charge moves by kinks rather than by rigid slides. What the framework does not deliver in any regime is a resistive conductivity, whose relaxation time needs a current as an object and irreversible dynamics: a minimisation over real densities has neither.

A Computational details

All calculations use the RealQM solver `molecule.js` driven by `chain_slide.html` (WebGPU, single GPU). Equation (1) is minimised by imaginary-time relaxation of the densities on a uniform Cartesian grid, with the partition specified by a cell-ownership map.

The code, and running it. Everything is at

`https://github.com/Claes542/RealMolecule`

and runs *in a web browser* with nothing to install, compile or configure: the solver is JavaScript and WGSL, the arithmetic is done by WebGPU on whatever GPU the machine has, and a run is started by opening a page. A WebGPU-capable browser is required — Chrome or Edge at present. Every configuration in this article is a query string, so each calculation below is reproduced by following one link. The reference state and the two windows that give the 240 meV of §4:

```
https://claes542.github.io/RealMolecule/chain_slide.html?n=16&a=4.5&rc=0&rsing=0.5&cps=8&
wrap=1&coax=1&ncore=2&chom=1&cr=2.4&d=0
```

and the same with `&swin=4&d=0.5`, `&swin=8&d=0.5`, `&swin=12&d=0.5`; the stiffened runs of §5 add `&kins=1`. Each run relaxes until its energy stops changing and then writes its total energy, its Euler–Lagrange residual and its per-domain eigenvalues into `localStorage` under `realqm_resid`, keyed by the full configuration, so the numbers in the tables below are read out rather than transcribed from the screen. The parameters are named in the source at the top of `chain_slide.html`. For every corrugation the partition is *frozen*: the energy of a stated configuration is evaluated, which is legitimate for configurations related by a symmetry of the lattice and is why translation by one spacing is used as the control. Energies are reported in hartree and converted at 1 Ha = 27211 meV.

The chain of hydrogen atoms. $N = 8$ atoms in a row at $a = 4.5\,a_0$, nuclear charge +1, one electron per site, $r_c = 0$. Wrapped labelling, so translation by one spacing is an exact relabelling. Mesh 16 cells per spacing, box $= (N-1)a + 10 = 41.5\,a_0$, grid 148^3, $h = 0.2804\,a_0$; nuclear cutoff $r_{\text{sing}} = 0.6\,a_0$, above $2h = 0.561$. Fixed 4000 steps.

$$E(0) = -3.624632386, \quad E(\tfrac{1}{2}) = -3.454076534, \quad E(1) = -3.624632386 \text{ Ha},$$

giving 580.1 meV per electron, with $E(1) = E(0)$ to nine digits as the control. Repeating at $r_{\text{sing}} = 0.8\,a_0$ (grid 148^3, same mesh) gives $E(0) = -3.395159382$ and $E(\frac{1}{2}) = -3.224867411$, i.e. 579.2 meV: the totals move by 0.23 Ha and the barrier by 0.9 meV. Polarisability from the induced dipole at $F = 5 \times 10^{-4}$ au with the partition frozen, at $N = 4, 6, 8$: $\alpha/N = 7.25,\ 7.43,\ 7.40\,a_0^3$.

The cable of Li ions. $N = 16$ sites at $a = 4.5\,a_0$ on the axis, kernel charge +3, one core domain per site carrying two electrons and confined within $c_r = 2.4\,a_0$ of the axis, one valence electron per site outside that radius: 32 domains, neutral, $r_c = 0$, so there is no pseudopotential and the carrier is excluded from the core region by non-overlap alone. Mesh 8 cells per spacing, box $= (N-1)a + 10 = 77.5\,a_0$, grid 138^3, $h = 77.5/138 = 0.5616\,a_0$; the box is the chain span plus a fixed pad and so does not depend on the mesh.

The nuclear cutoff, and a correction. A bare $-Z/r$ well cannot be represented on a grid, so it is clamped at a radius R_{sing}. The solver sets $R_{\text{sing}} = \max(2h, r_{\text{sing}})$ with r_{sing} a requested absolute value, and the requested $0.5\,a_0$ used here is smaller than $2h$ at every mesh employed — 1.12, 0.90 and $0.75\,a_0$ at 8, 10 and 12 cells per spacing. The cutoff was therefore $2h$ throughout, tied to the grid rather than absolute, and refining the mesh sharpens the ion instead of merely resolving it. An earlier version of this appendix stated that the clamp fixed the potential as the same object at every mesh; that is not what the code does, and any comparison across meshes made under that assumption compares different ions rather than one ion at different resolutions. To hold the potential fixed, r_{sing} must be set above $2h$ at the finest mesh in use. Only the shell slides. All runs use the fixed-step protocol below, at $r_{\text{sing}} = 1.5\,a_0$ (above $2h = 1.123$). Windowed energies at $d = \frac{1}{2}$, W the number of displaced carriers:

	$W = 4$	$W = 8$
E (Ha), 4000 steps	−149.411042	−149.377292
E (Ha), stiffened, 8000 steps	−157.956523	−157.961137

Equation (3) gives $\varepsilon_{\text{bulk}} = +229.6$ meV per carrier unstiffened and -31.4 meV stiffened.

Stiffened carriers. Identical geometry, mesh and ion lattice, with the coefficient of $|\nabla\psi_i|^2$ raised from $\frac{1}{2}$ to 1 for the shell domains only, the cores keeping $\frac{1}{2}$. The coefficient enters three places — the imaginary-time step, the $H\psi$ diagnostic and the energy sum — and all three must carry it or the reported energy does not belong to the state produced. The explicit step requires the diffusion number below 1/6, so the timestep is halved in proportion, and the stiffened runs are given 8000 steps rather than 4000 to cover the same relaxation.

The fixed-step protocol. Runs are stopped after a stated number of relaxation steps rather than when the energy stops changing. The energy test — three consecutive frames within 2×10^{-5} Ha — halts at a different step count for every configuration, so two windowed energies would be relaxed to different degrees and their difference, which is the quantity wanted, would mix the two; and a configuration that never satisfies it runs to the step cap, which is weeks. A fixed count makes each run reproducible from its URL, the solver being deterministic to nine digits, and makes the windowed difference a comparison of like with like. Calibration on the cable at $c_r = 2.4$, $M = 4$: $E = -155.268964$ at 200 steps, -150.144412 at 1000, -149.483339 at 2000, -149.411042 at 4000, -149.410862 at 8000. Four thousand steps is used throughout; eight thousand moves the energy by 4.9 meV. The Euler–Lagrange residual plateaus at 4.6×10^{-3} from 2000 steps onward while the energy is still 2 eV out at that point, so the residual detects gross non-convergence but cannot be the sole test.

The rate estimate of §6. Equation (4) evaluated at $T = 300$ K ($kT = 25.9$ meV) with $a = 4.5\,a_0 = 2.381$ Å, $n = 1/a^3 = 7.4 \times 10^{28}$ m^{-3}, the nuclear mass of the moving species, and $K = 2\pi^2 V_0/a^2$ from the assumed sinusoid, giving $\nu = (1/a)\sqrt{V_0/2M}$:

V_0 (meV)	K (eV Å^{-2})	ν (Hz)	σ_0 (S/m)	$e^{-V_0/kT}$	σ (S/m)
1000	3.48	5.5×10^{12}	1.4×10^5	1.6×10^{-17}	2.3×10^{-12}
240	0.84	2.7×10^{12}	7.0×10^4	9.3×10^{-5}	6.5
24	0.084	8.5×10^{11}	2.2×10^4	0.395	8.8×10^3

Taking $M = m_e$ instead of an ionic mass raises ν to 6.1×10^{14} Hz and σ to 1.5×10^3 S m^{-1} on the middle row; that frequency is two orders above any lattice vibration, which is the argument for the ionic choice quite apart from the physical one. The ± 30 meV uncertainty on the middle row is a factor 3.2 in σ.

Convergence and resolution. The states entering each windowed difference are relaxed but not fully converged, carrying Euler–Lagrange residuals of about 4×10^{-3}; the stiffened runs halve the timestep for stability and are less relaxed than the others. Every number reported is for the stated system at the stated mesh, with the nuclear cutoff held at an absolute radius above $2h$ so that it belongs to the model rather than to the grid. Whether the same numbers survive a different discretisation, or a different implementation of the same equations, is not established here and is left to further work.

Controls applied. Translation by one full period must return the same energy: satisfied to nine digits ($E(0) = E(N)$ exactly) on the wrapped geometries. Repeat runs of one configuration agree to nine digits, so the solver is deterministic and any discrepancy is attributable to the model or the input. Domains related by a symmetry must have equal eigenvalues: this is what exposed the terminal domains of a finite cable binding an electron-volt harder than the interior, and hence the necessity of (3). Euler–Lagrange residuals are 4×10^{-3} for the chain and $2–3 \times 10^{-2}$ for the cable.

On the use of computational assistance

The solver and the drivers used here were written with AI assistance, and the analysis was carried out the same way. Two statements make that disclosure precise, and the second is what licenses the first.

No quantity in the model is calibrated to any measurement. (1) contains Coulomb interaction and kinetic energy and nothing else; what the calculation takes beyond it is a specification of the system — which nuclei, at what spacing, under what geometric constraint — and a stated convention for reporting a bulk density. Nothing was adjusted to reproduce a barrier, a conductivity or any other number, and none of the values reported was known in advance of computing it.

The implementation is checkable against standards independent of it, and was checked. A hydrogen atom must come out at -0.5 Ha; two separated unit charges must repel as $1/R$; translation of a wrapped chain by one spacing is a relabelling and must return the same energy, which it does to nine digits; repeat runs of one configuration must agree, and do, to the last reported digit. That discipline is not a formality. It found, in the course of this work, that the nuclear cutoff was tied to the mesh rather than fixed, so that refining the grid sharpened the ion instead of resolving it; and that nuclear coordinates are truncated to whole cells before the solver sees them, which is why several quantities named in §9 cannot presently be computed at all. Neither is a fitted parameter, and both changed what the calculation means.

The distinction worth drawing is therefore not between assistance with code and assistance with physics, which would suppose that implementation is innocent — it is not, as those two findings show. It is between content calibrated to the data it is meant to explain, which absorbs disagreement, and content derived from a stated principle, which is exposed to it. This article contains none of the first kind.

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